

MITHIL K GOWDA

☎ +91-8431196506 ✉ mithil.k.g.10@gmail.com 🔗 LinkedIn 🐙 GitHub 🌐 Portfolio

Professional Summary

Cybersecurity graduate specializing in Cloud Security, Cloud Infrastructure Security, Identity and Access Management (IAM), Cloud Security Architecture, Secure Cloud Deployment, and Secure Software Engineering. Project Intern at ISRO–LEOS with experience developing secure backend applications using Python, Flask, REST APIs, and MySQL. First author of internationally recognized cybersecurity research focused on secure cloud-ready architectures, AI-driven cyber defense, cryptographic security, and resilient distributed systems. Passionate about designing secure cloud-native applications and implementing cloud governance, identity security, and enterprise cloud protection. Seeking graduate opportunities in Cloud Security Engineering, Cloud Infrastructure Security, and Enterprise Cloud Solutions.

Education

M.S. Ramaiah University of Applied Sciences, Bengaluru

2022–2026

B.Tech. in Information Science and Engineering

CGPA: **8.80** / 10

Technical Skills

Cloud Security: Google Cloud Platform (GCP), Cloud Infrastructure Security, Cloud Security Architecture, Secure Cloud Deployment, Identity and Access Management (IAM), Cloud Identity, Cloud Governance, Shared Responsibility Model, Cloud Security Best Practices

Cloud Concepts: Cloud Networking, High Availability, Scalability, Fault Tolerance, Disaster Recovery, Load Balancing

Application Security: Authentication, Authorization, API Security, Secure Software Development, Security Architecture, OWASP Top 10

Programming: Python, Java, JavaScript, SQL, HTML, CSS, Bash

Backend Development: Flask, REST APIs, Backend Application Development, API Design

Databases: MySQL, MongoDB

Networking: TCP/IP, DNS, HTTP/HTTPS, SSH, VPN, Network Security, TLS/SSL

Security Tools: Burp Suite, OWASP ZAP, Nmap, Wireshark, Postman, Git, GitHub

Cryptography: AES-256, SHA-256, Zero-Knowledge Proofs (ZKP), Secure Transaction Verification

Operating Systems: Linux, Windows

Professional Experience

Indian Space Research Organisation (ISRO) – Laboratory for Electro Optics Systems (LEOS), Bengaluru
2026

Project Intern – Secure Software Development & Data Analytics

- Designed and developed secure backend applications using Python, Flask, REST APIs, and MySQL following cloud-native application security and secure software engineering principles.
- Implemented authentication, authorization, secure API communication, database security controls, and identity-aware application workflows supporting cloud-ready architectures.
- Developed scalable backend services emphasizing modular architecture, secure cloud deployment readiness, secure data processing, and enterprise application reliability.
- Engineered maintainable backend components supporting secure software architecture, cloud scalability, secure communication, and resilient enterprise application development.

Cloud Security & Distributed Systems Projects

A Multi-Agent AI Cyber Defense Framework Using HoneyBee Method

- First Author and Corresponding Author of an AI-driven cybersecurity framework accepted for presentation at ICIICE 2026 (Dubai, UAE) and accepted as a book chapter in the edited volume *Metaheuristic Optimization for Social Good*.

- Designed a cloud-ready distributed security architecture integrating intelligent threat detection, adaptive cyber defense, distributed threat intelligence, secure cloud services, and resilient enterprise cloud security.
- Developed cloud-compatible machine learning security services integrating XGBoost, CNN-LSTM, Deep Q-Network (DQN), adaptive honeypots, distributed security intelligence, and cloud-native security principles.
- Evaluated the framework using CIC-IDS2017 datasets and simulated enterprise cloud attack scenarios, achieving 98.24% detection accuracy, 98.51% precision, 98.10% recall, and 0.992 ROC-AUC.

AI-Governed Carbon Credit Exchange with Digital Carbon Passport

- Abstract accepted at the 4th International Conference on Advancing Sustainable Futures (ICASF 2027), Abu Dhabi University, UAE.
- Designed a scalable cloud-ready distributed application integrating Digital Carbon Passports, cloud identity management, authentication, authorization, trust scoring, fraud detection, and secure transaction management.
- Developed the Secure Trade Authorization and Verification Pipeline (STAVP) utilizing AES-256 encryption, SHA-256 hashing, secure authentication, API protection, cloud-ready transaction validation, and cryptographic security controls.

C³T-STAVP Cryptographic Framework (Ongoing Research)

- Designing a cloud-compatible cybersecurity framework integrating Character Chaffing & Transposition Technology (C³T), Secure Transaction Authorization & Verification Protocol (STAVP), and adaptive trust validation.
- Researching Zero-Knowledge Proofs (ZKP), post-quantum cryptography, secure distributed architectures, semantic obfuscation, cloud-ready security mechanisms, and resilient transaction validation.
- Developing cryptographic security mechanisms supporting secure cloud applications, distributed trust management, cloud-native security architectures, enterprise-scale security, and resilient cloud services.

Professional Certifications

- Google Cloud Computing – NPTEL
- IBM IT Fundamentals for Cybersecurity Specialization – IBM
- Ethical Hacker – Cisco Networking Academy
- Systems and Usable Security (Elite) – NPTEL
- Introduction to Cyber Security – IGNOU (SWAYAM)
- IBM Front-End Developer Specialization – IBM
- Introduction to Internet of Things (Silver Medal) – NPTEL
- Business Intelligence and Analytics – NPTEL

Achievements

- Selected as Project Intern at the Indian Space Research Organisation (ISRO – LEOS).
- First Author of internationally recognized research focused on AI-driven distributed systems, intelligent software architectures, and cybersecurity.
- Awarded the Reliance Foundation Undergraduate Scholarship (2026).
- Recipient of NPTEL Silver Medal in Introduction to Internet of Things.